

Homework 4

Due February 15th

Problems:

1. Book Problem 2.43: Without substantial work, the operation of OA3 is unknown as it does not clearly have negative feedback; however, you may assume that it still operates such that virtual short analysis is still applicable (which it is as negative feedback is provided by OA2 within the loop). Further, although the problem states the wiper of R_1 is “nominally positioned halfway,” you may assume that it is indeed positioned at the halfway point. R_1 is used to “trim” the circuit; accounting for component tolerances to ensure equal gain between both V_1 and V_2 .
2. Book Problem 2.47: Assume low-frequency operation.
3. Book Problem 2.58: Assume a 1000 Ω Pt RTD ([example resistance-vs-temperature table](#)). A Pt1000 RTD has a sensitivity of approximately 0.00385 $\Omega/\Omega/^\circ\text{C}$ (DIN 43760 Standard). Also assume that this is for a low-power implementation (hence the PT1000 RTDs), therefore target a current of 0.1 mA through each RTD. How much power is being consumed by all the resistors in the circuit?